

EXPERIMENT 18

Edge Detection Using Sobel Matrix Along Y-Axis

Aim

To perform edge detection using the Sobel matrix along the Y-axis on an input image using Python and OpenCV.

Procedure

1. Import the required libraries, cv2 and numpy.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply the Sobel operator along the Y-axis to detect horizontal edges.
5. Convert the result into an appropriate display format.
6. Display the original image and the Sobel Y edge-detected image.
7. Save the output image.

Sobel Y Matrix

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Input



Output



Result

Thus, edge detection along the Y-axis was successfully performed using the Sobel operator, and the horizontal edges in the image were detected.